

# ERA 5 calibration to Elexon power

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2026-06-11

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## Overview

# Updates

## New

- Error diagnostics
- Spatiotemporal model for calibration

## Ongoing

- Model validation
  - Out-of-sample in time
  - Out-of-sample in space
- Consistency under aggregation
- Low wind events



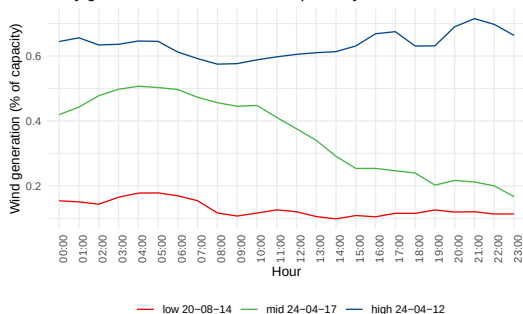
## Model performance assessment

# Assessment framework

- Run models on a sample of days
- Performance metrics:
  - RMSE
  - MAE
  - Bias
- Evaluate performance at WF level
- Error under aggregation
  - GB level
  - Onshore / offshore
  - By regime
  - By hour

## Sampled days for performance assessment

Daily generation time series for sampled days



# Model portfolio

- Wind farm level models:
  - Generic Power Curve (ERA5 + generic PC estimate)
  - Quantile Mapping
  - Linear model
  - Spatiotemporal model
- Linear model at Great Britain (GB) level

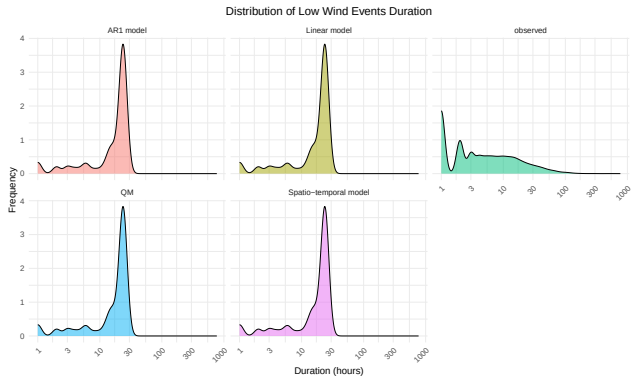
## Performance under different regimes

# Error metrics by regime

| regime | model                 | RMSE  | MAE   | MDAPE  | Bias      |
|--------|-----------------------|-------|-------|--------|-----------|
| Low    | Generic PC            | 0.157 | 0.078 | 85.417 | 4.60e-02  |
| Low    | QM                    | 0.122 | 0.056 | 96.021 | -1.20e-04 |
| Low    | GB LM                 | 0.116 | 0.063 | 76.195 | 7.60e-03  |
| Low    | Linear model          | 0.111 | 0.056 | 75.148 | 2.06e-03  |
| Low    | Spatio-temporal model | 0.029 | 0.014 | 28.703 | 1.17e-03  |
| Low    | AR2 model             | 0.002 | 0.001 | 2.306  | 2.41e-05  |
| Low    | AR1 model             | 0.000 | 0.000 | 0.006  | 5.33e-08  |
| High   | Generic PC            | 0.388 | 0.298 | 34.078 | 2.68e-01  |
| High   | QM                    | 0.324 | 0.245 | 31.613 | 9.82e-05  |
| High   | GB LM                 | 0.270 | 0.213 | 23.829 | 3.88e-02  |
| High   | Linear model          | 0.245 | 0.197 | 24.649 | 2.29e-04  |
| High   | AR2 model             | 0.153 | 0.120 | 15.307 | 5.80e-05  |
| High   | AR1 model             | 0.141 | 0.111 | 14.058 | 2.26e-05  |
| High   | Spatio-temporal model | 0.121 | 0.088 | 11.285 | 2.78e-04  |
| Mid    | Generic PC            | 0.281 | 0.194 | 52.151 | 1.48e-01  |
| Mid    | QM                    | 0.203 | 0.136 | 45.336 | 3.35e-05  |
| Mid    | GB LM                 | 0.188 | 0.131 | 35.379 | 1.99e-02  |
| Mid    | Linear model          | 0.170 | 0.120 | 35.243 | 2.57e-03  |
| Mid    | Spatio-temporal model | 0.094 | 0.062 | 19.867 | 2.63e-03  |
| Mid    | AR2 model             | 0.073 | 0.051 | 16.905 | 9.31e-04  |
| Mid    | AR1 model             | 0.071 | 0.050 | 16.218 | 1.05e-03  |

# Low wind speed events

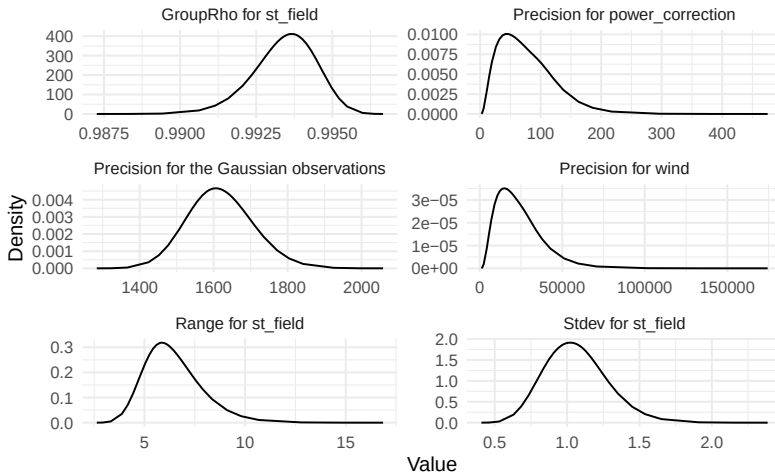
## Low wind speed events distribution



Low wind speed

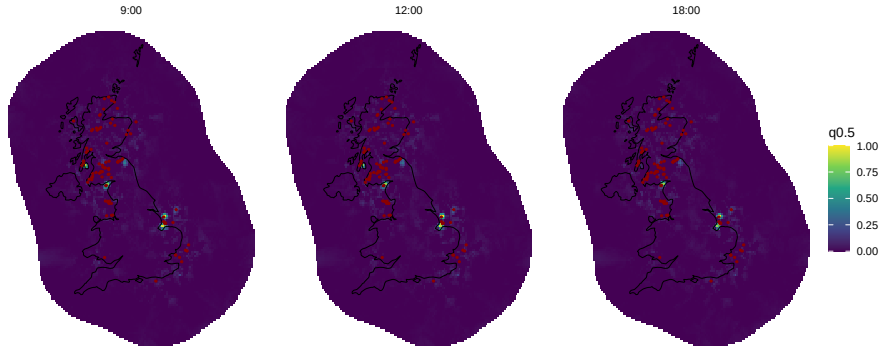
# Hyperparameters

## Density of Hyperparameters

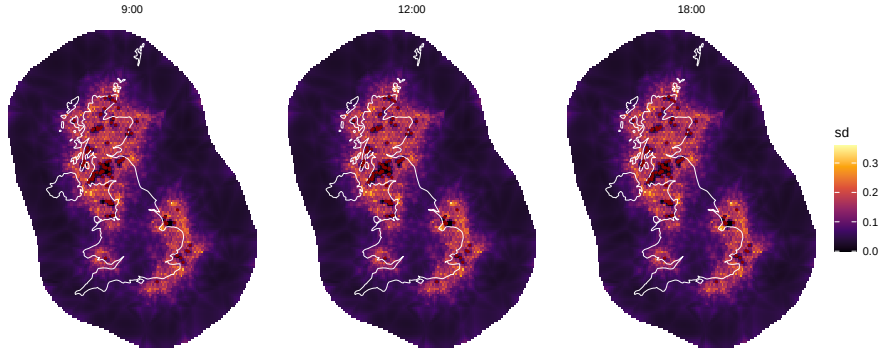




# Estimated spatial field - median



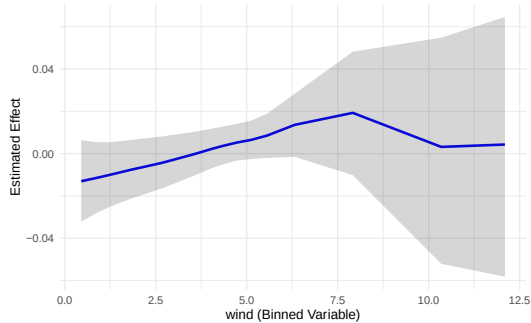
# Estimated spatial field - standard deviation



# Effect of wind speed and power regime

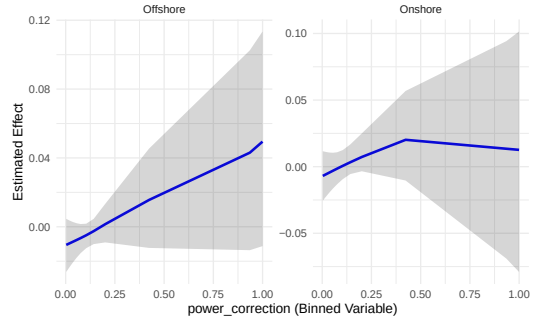
## Wind speed effect

Estimated effect for wind



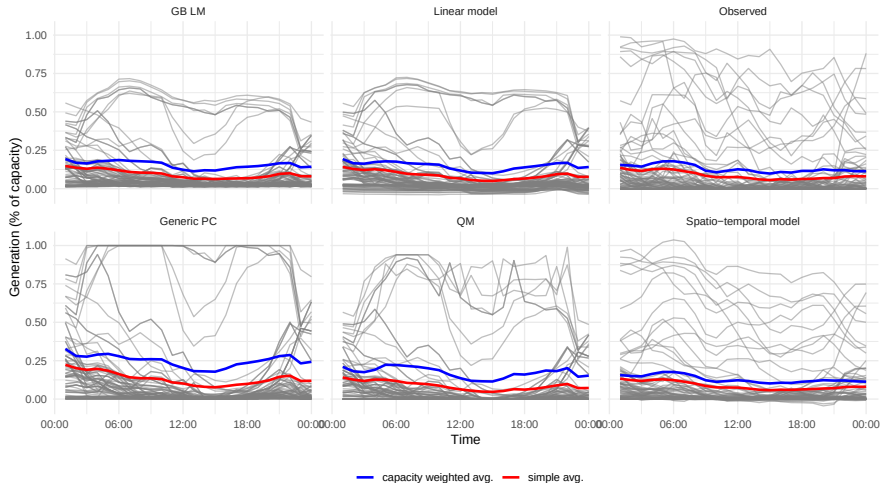
## Power regime effect

Estimated effect for power\_correction



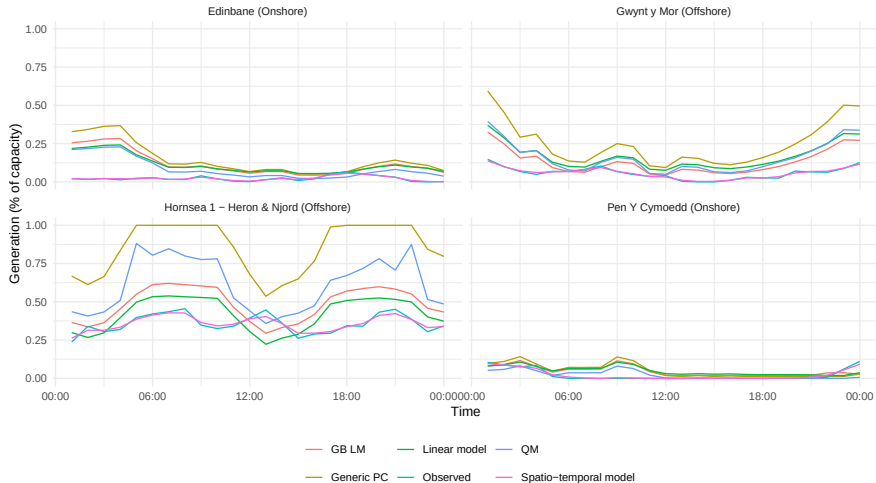
# Power estimates

Power estimates Time Series 2020-08-14



# Power estimates sampled wind farms

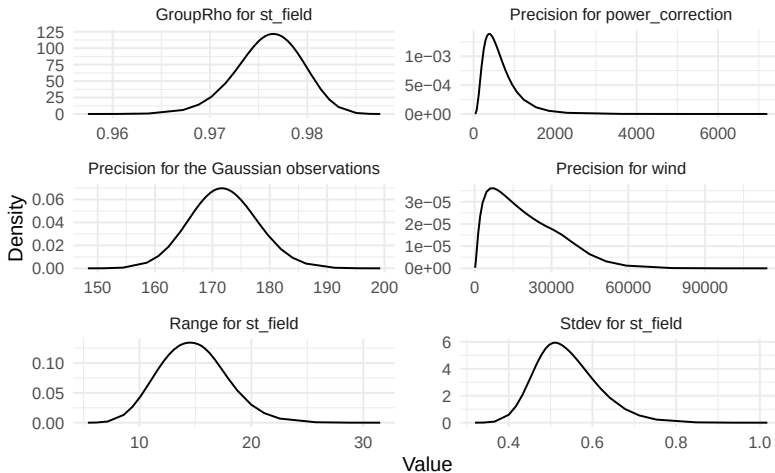
Power estimates Time Series 2020-08-14



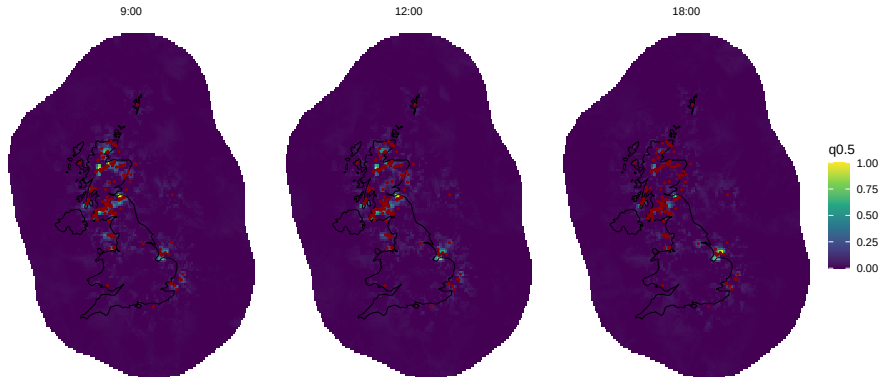
Medium wind speed

# Hyperparameters

## Density of Hyperparameters

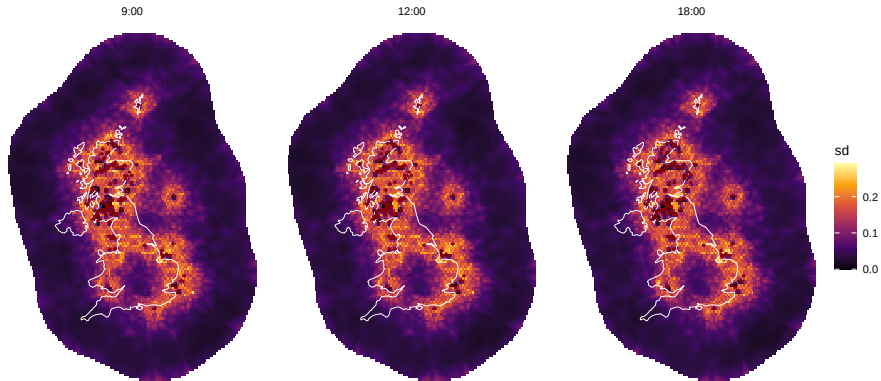


# Estimated spatial field - median





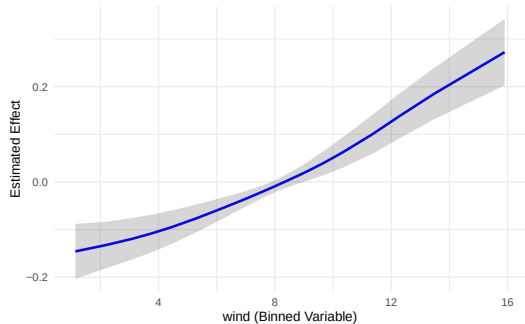
# Estimated spatial field - standard deviation



# Effect of wind speed and power regime

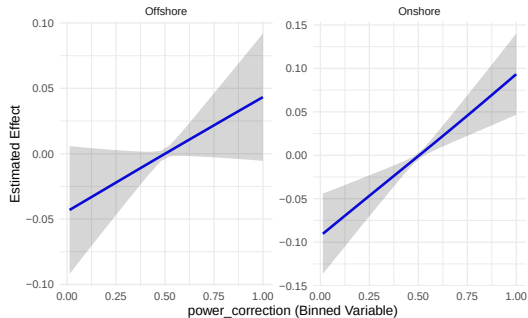
## Wind speed effect

Estimated effect for wind



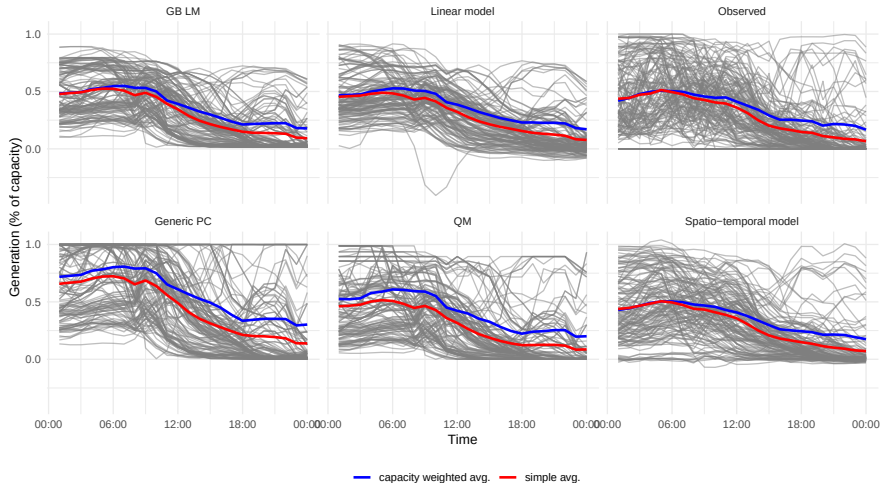
## Power regime effect

Estimated effect for power\_correction



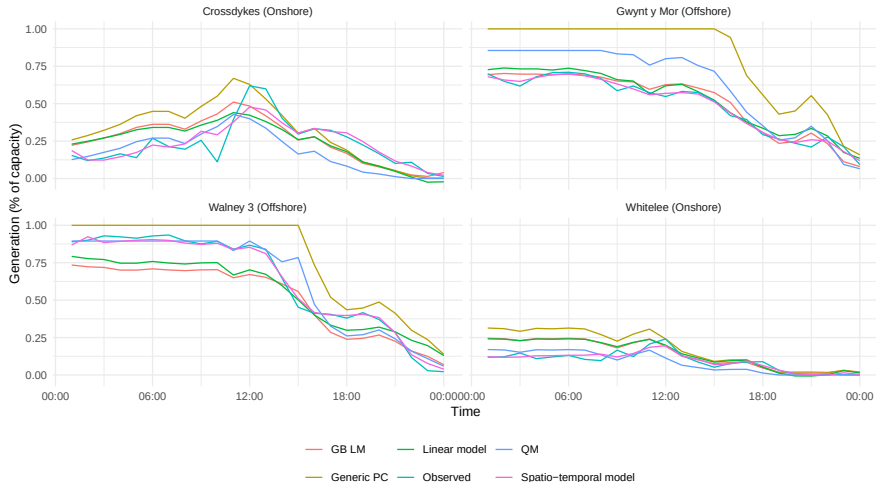
# Power estimates

Power estimates Time Series 2024-04-17



# Power estimates sampled wind farms

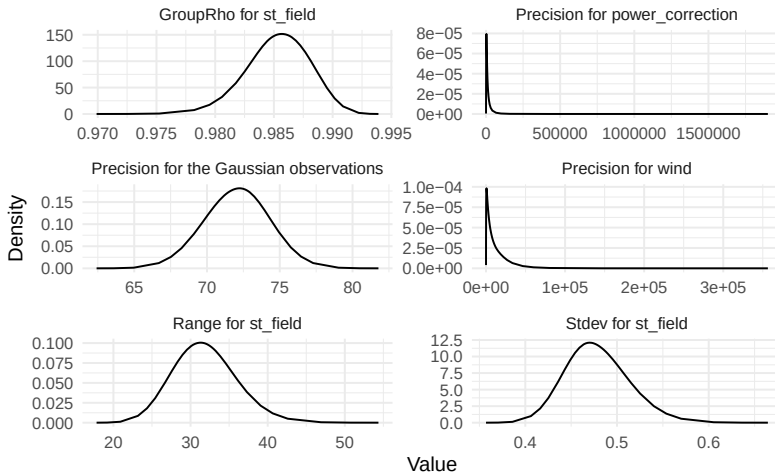
Power estimates Time Series 2024-04-17



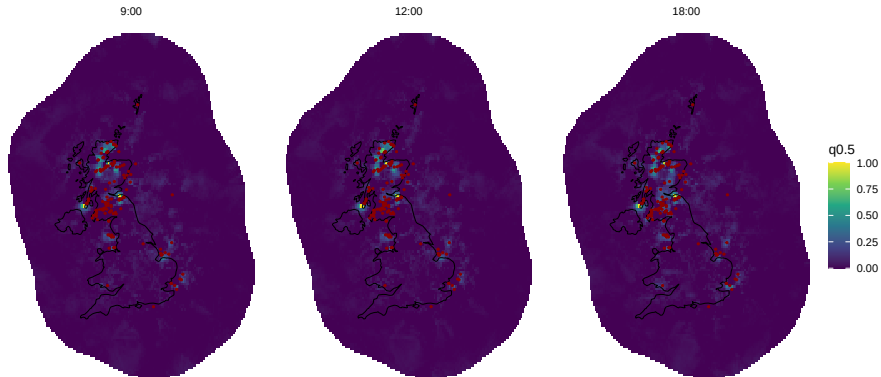
High wind speed

# Hyperparameters

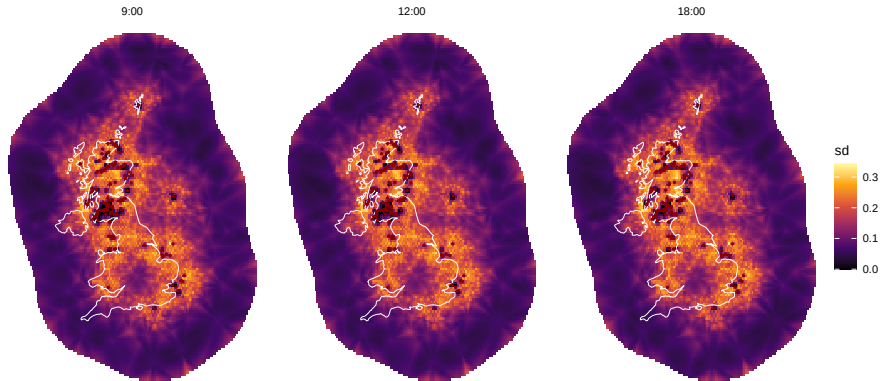
Density of Hyperparameters



# Estimated spatial field - median



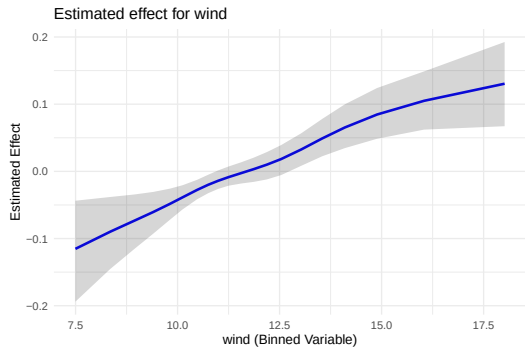
# Estimated spatial field - standard deviation





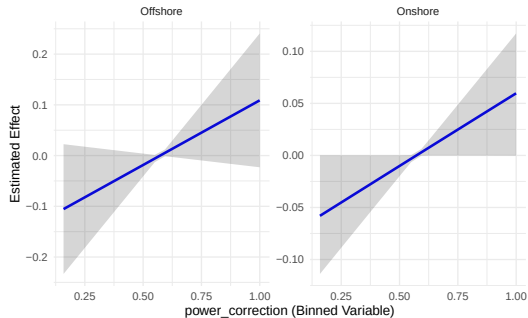
# Effect of wind speed and power regime

## Wind speed effect



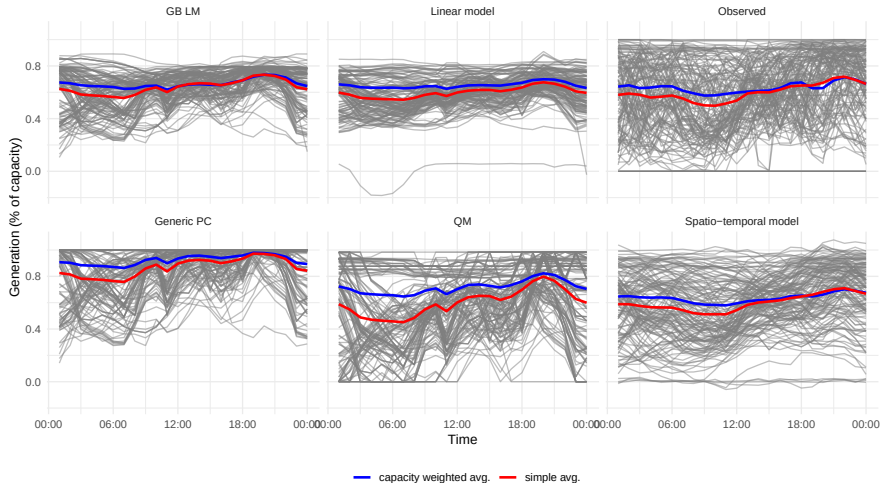
## Power regime effect

Estimated effect for power\_correction

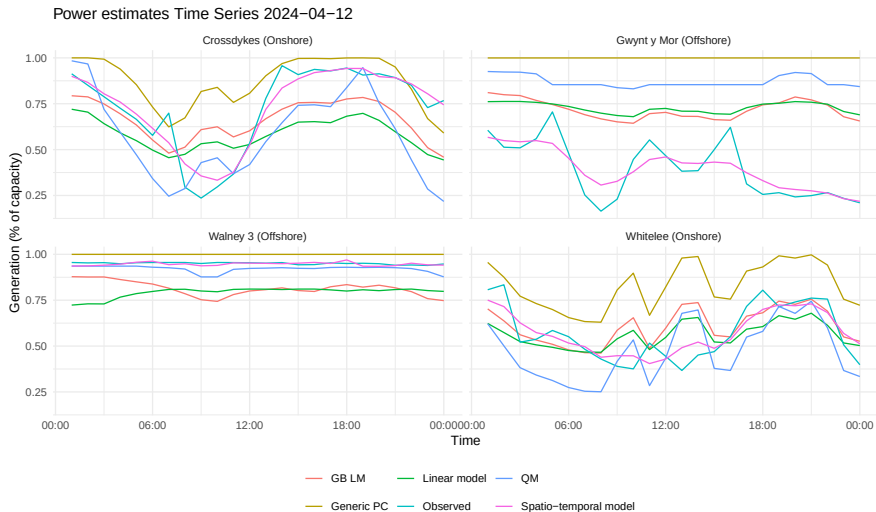


# Power estimates

Power estimates Time Series 2024-04-12



# Power estimates sampled wind farms



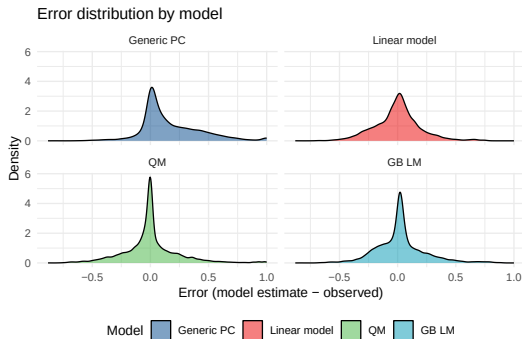
August 10th, 2024

# Error metrics

## Metrics at wind farm level

| model        | RMSE  | MAE   | Bias      |
|--------------|-------|-------|-----------|
| Generic PC   | 0.305 | 0.206 | 1.69e-01  |
| QM           | 0.235 | 0.153 | 1.14e-04  |
| GB LM        | 0.211 | 0.146 | 2.42e-02  |
| Linear model | 0.198 | 0.144 | -3.25e-15 |

## Error distribution by model

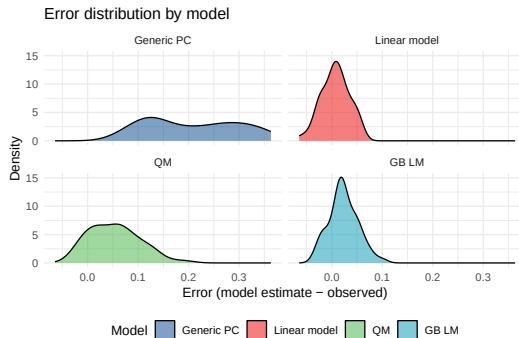


# Error metrics aggregated

Metrics at GB level

| model        | RMSE  | MAE   | Bias     |
|--------------|-------|-------|----------|
| Generic PC   | 0.225 | 0.207 | 2.07e-01 |
| QM           | 0.073 | 0.058 | 5.33e-02 |
| GB LM        | 0.037 | 0.030 | 2.32e-02 |
| Linear model | 0.028 | 0.023 | 7.19e-03 |

Error distribution by model



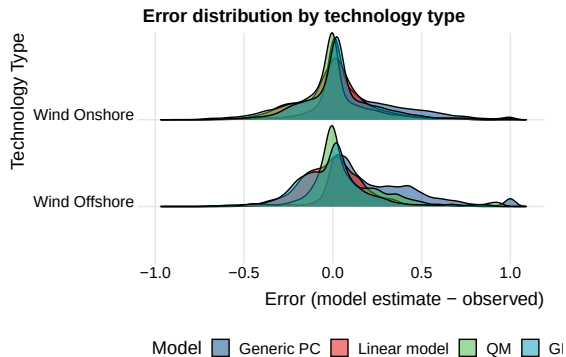
# Technology type

Metrics by technology type

| tech_typ      | model        | RMSE  | MAE   | Bias      |
|---------------|--------------|-------|-------|-----------|
| Wind Offshore | Generic PC   | 0.350 | 0.253 | 2.34e-01  |
| Wind Offshore | QM           | 0.253 | 0.160 | 8.46e-02  |
| Wind Offshore | GB LM        | 0.215 | 0.149 | -2.19e-03 |
| Wind Offshore | Linear model | 0.193 | 0.143 | -2.21e-14 |
| Wind Onshore  | Generic PC   | 0.287 | 0.189 | 1.46e-01  |
| Wind Onshore  | QM           | 0.228 | 0.150 | -3.05e-02 |
| Wind Onshore  | GB LM        | 0.210 | 0.144 | 3.37e-02  |
| Wind Onshore  | Linear model | 0.200 | 0.144 | 3.61e-15  |

# Technology type

Error distribution by technology type



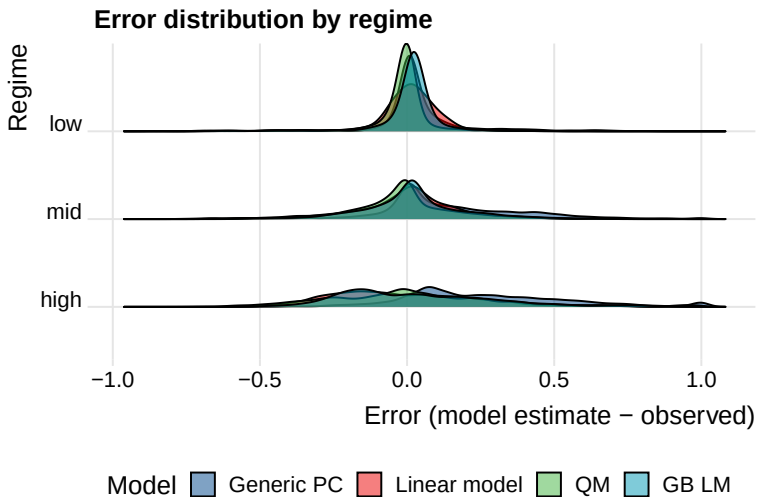


# Regime

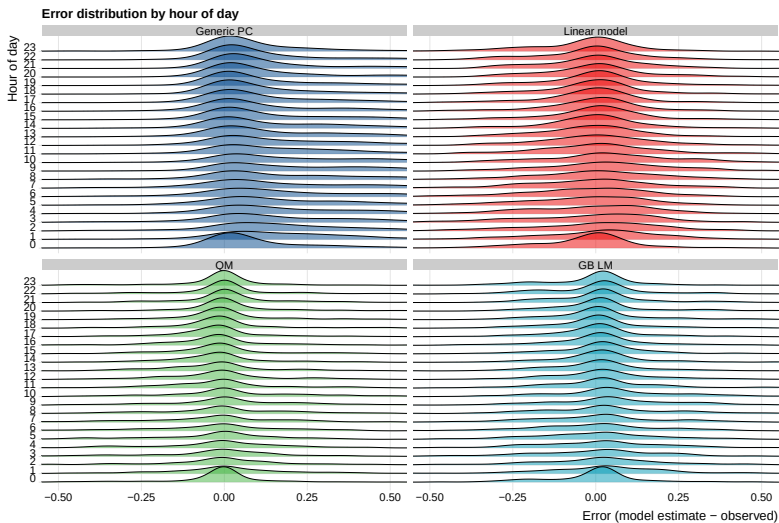
Metrics by regime

| p_group3 | model        | RMSE  | MAE   | Bias      |
|----------|--------------|-------|-------|-----------|
| low      | Generic PC   | 0.157 | 0.078 | 4.60e-02  |
| low      | QM           | 0.124 | 0.054 | -1.31e-02 |
| low      | Linear model | 0.120 | 0.076 | 5.47e-03  |
| low      | GB LM        | 0.116 | 0.063 | 7.60e-03  |
| mid      | Generic PC   | 0.281 | 0.194 | 1.48e-01  |
| mid      | QM           | 0.209 | 0.140 | -1.84e-02 |
| mid      | GB LM        | 0.188 | 0.131 | 1.99e-02  |
| mid      | Linear model | 0.177 | 0.127 | 5.26e-03  |
| high     | Generic PC   | 0.388 | 0.298 | 2.68e-01  |
| high     | QM           | 0.303 | 0.228 | 2.70e-02  |
| high     | GB LM        | 0.270 | 0.213 | 3.88e-02  |
| high     | Linear model | 0.250 | 0.204 | -8.70e-03 |

# Regime



# Hour of the day

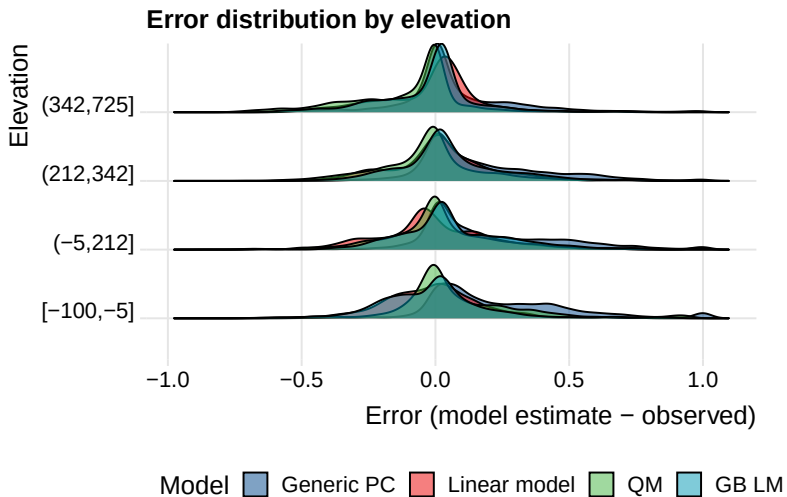


# Elevation

Metrics by regime

| elevation_g4 | model        | RMSE  | MAE   | Bias      |
|--------------|--------------|-------|-------|-----------|
| [-100,-5]    | Generic PC   | 0.353 | 0.257 | 2.37e-01  |
| [-100,-5]    | QM           | 0.256 | 0.163 | 8.64e-02  |
| [-100,-5]    | GB LM        | 0.217 | 0.151 | -1.74e-03 |
| [-100,-5]    | Linear model | 0.194 | 0.144 | -3.74e-03 |
| (-5,212]     | Generic PC   | 0.330 | 0.227 | 1.97e-01  |
| (-5,212]     | QM           | 0.234 | 0.154 | 2.20e-02  |
| (-5,212]     | GB LM        | 0.228 | 0.162 | 6.81e-02  |
| (-5,212]     | Linear model | 0.214 | 0.160 | -4.95e-03 |
| (212,342]    | Generic PC   | 0.292 | 0.199 | 1.71e-01  |
| (212,342]    | QM           | 0.212 | 0.143 | -1.44e-02 |
| (212,342]    | GB LM        | 0.202 | 0.141 | 5.18e-02  |
| (212,342]    | Linear model | 0.192 | 0.138 | 2.27e-02  |
| (342,725]    | QM           | 0.236 | 0.152 | -9.77e-02 |
| (342,725]    | Generic PC   | 0.227 | 0.139 | 6.93e-02  |
| (342,725]    | GB LM        | 0.194 | 0.128 | -2.06e-02 |
| (342,725]    | Linear model | 0.191 | 0.134 | -1.35e-02 |

# Elevation

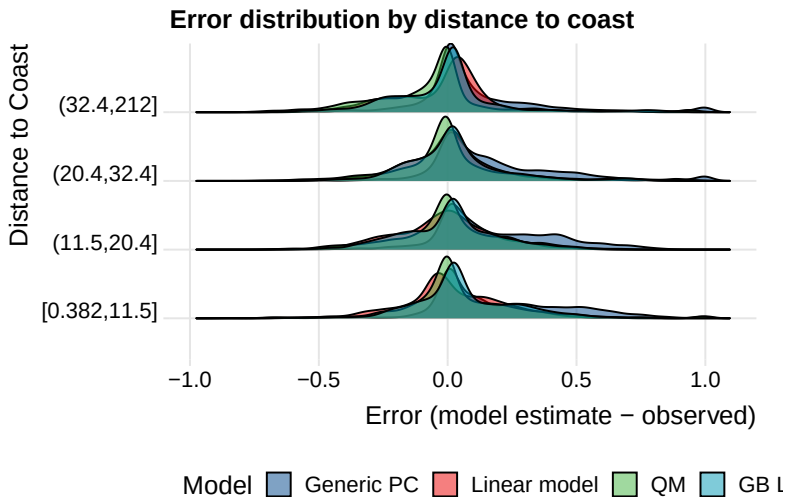


# Distance to coast

Metrics by distance to coast

| dist_coast_g4 | model        | RMSE  | MAE   | Bias      |
|---------------|--------------|-------|-------|-----------|
| [0.382,11.5]  | Generic PC   | 0.335 | 0.237 | 2.06e-01  |
| [0.382,11.5]  | QM           | 0.233 | 0.150 | 4.77e-02  |
| [0.382,11.5]  | GB LM        | 0.217 | 0.153 | 6.84e-02  |
| [0.382,11.5]  | Linear model | 0.199 | 0.149 | 1.66e-02  |
| (11.5,20.4]   | Generic PC   | 0.297 | 0.217 | 1.86e-01  |
| (11.5,20.4]   | QM           | 0.210 | 0.147 | 8.37e-03  |
| (11.5,20.4]   | GB LM        | 0.197 | 0.146 | 7.72e-03  |
| (11.5,20.4]   | Linear model | 0.189 | 0.145 | -1.45e-02 |
| (20.4,32.4]   | Generic PC   | 0.311 | 0.208 | 1.79e-01  |
| (20.4,32.4]   | QM           | 0.237 | 0.149 | 1.14e-02  |
| (20.4,32.4]   | GB LM        | 0.205 | 0.138 | 2.92e-02  |
| (20.4,32.4]   | Linear model | 0.198 | 0.136 | 1.30e-02  |
| (32.4,212]    | Generic PC   | 0.274 | 0.162 | 1.06e-01  |
| (32.4,212]    | QM           | 0.258 | 0.166 | -6.78e-02 |
| (32.4,212]    | GB LM        | 0.224 | 0.146 | -9.56e-03 |
| (32.4,212]    | Linear model | 0.205 | 0.147 | -1.52e-02 |

# Distance to coast



## Spatiotemporal calibration model

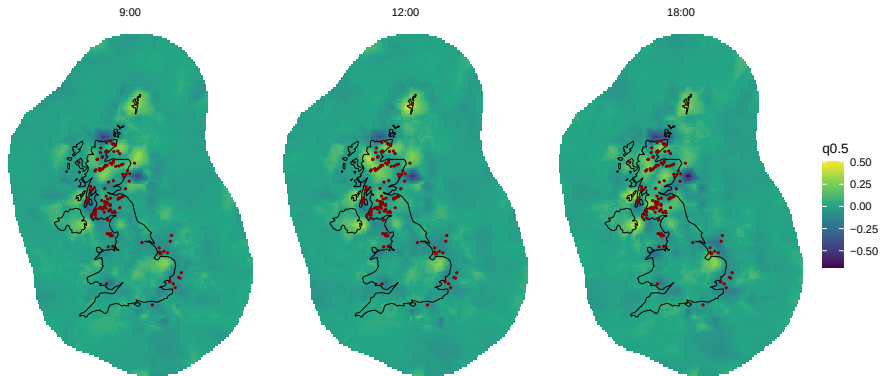


# Model overview

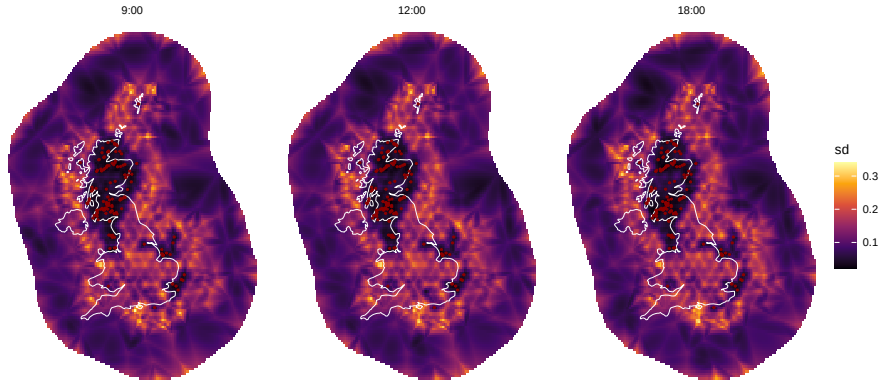
$$\mathbb{E}(p_{it}^{\text{calib}}) = \alpha + \beta_k + \beta_m + \beta_h + s_p(\hat{p}_{it}) + s_w(w_{it}) + u_{it}, \quad (1)$$

- Calibrated power: corrected output at site  $i$ , time  $t$ .
- $\alpha$ : intercept.
- Fixed effects  $\beta_k, \beta_m, \beta_h$ : location/season/time structure.
- ERA5 correction  $s_p$ : bias adjustment of ERA5 + generic PC estimate.
- Wind effect  $s_w$ : nonlinear wind–power correction.
- Spatiotemporal effect  $u_{it}$ : Spatial component + AR1 error.

# Estimated spatial field



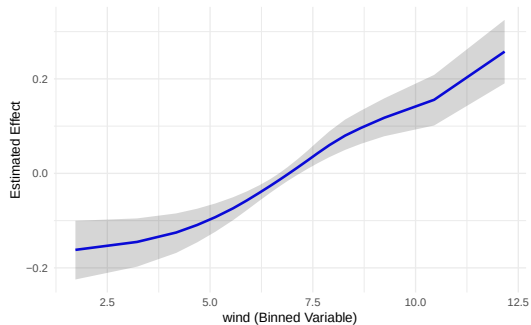
# Standard deviation of the spatial field



# Random Effects

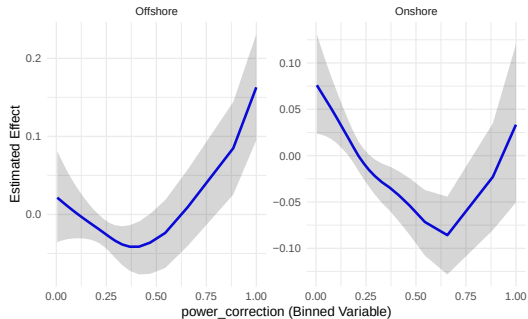
## Correction by wind speed

Estimated effect for wind



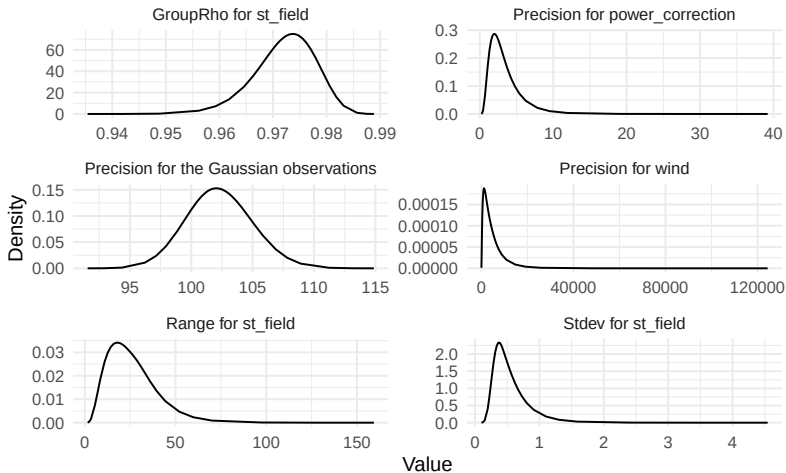
## Correction by power regime

Estimated effect for power\_correction



# Hyperparameters

## Density of Hyperparameters



## Comparison scores

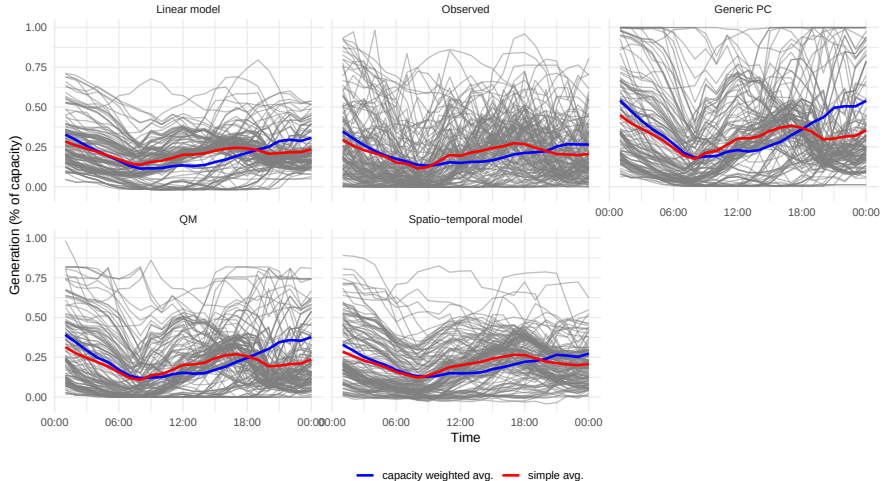
| model                 | RMSE  | MAE   | Bias      |
|-----------------------|-------|-------|-----------|
| Generic PC            | 0.213 | 0.153 | 1.02e-01  |
| QM                    | 0.154 | 0.108 | -2.50e-04 |
| Linear model          | 0.137 | 0.101 | -1.15e-17 |
| Spatio-temporal model | 0.094 | 0.065 | 8.57e-11  |

## Scores by Offshore / Onshore

| tech_typ      | model                 | RMSE  | MAE   | Bias      |
|---------------|-----------------------|-------|-------|-----------|
| Wind Offshore | Generic PC            | 0.240 | 0.161 | 1.44e-01  |
| Wind Offshore | QM                    | 0.142 | 0.087 | 4.60e-02  |
| Wind Offshore | Linear model          | 0.112 | 0.077 | 4.33e-16  |
| Wind Offshore | Spatio-temporal model | 0.065 | 0.040 | -3.78e-03 |
| Wind Onshore  | Generic PC            | 0.204 | 0.151 | 8.77e-02  |
| Wind Onshore  | QM                    | 0.159 | 0.114 | -1.58e-02 |
| Wind Onshore  | Linear model          | 0.144 | 0.109 | -1.61e-16 |
| Wind Onshore  | Spatio-temporal model | 0.101 | 0.073 | 1.27e-03  |

# Estimates by hour

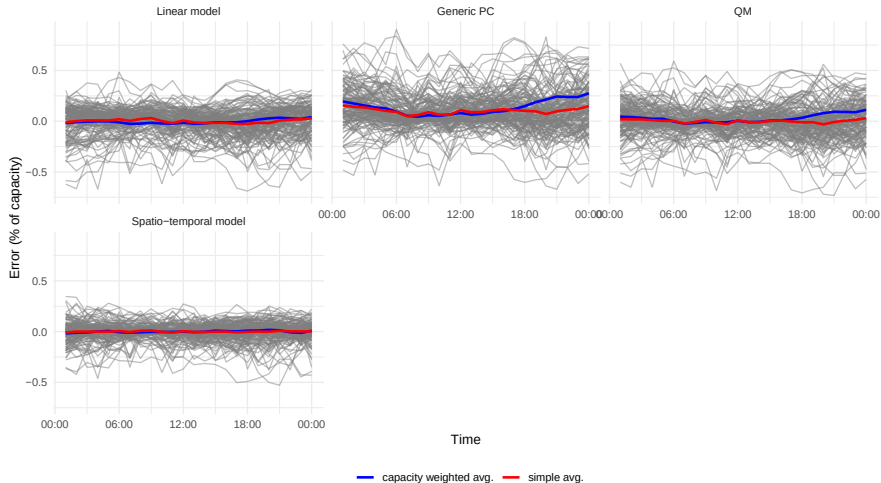
Power estimates Time Series 2025-08-10





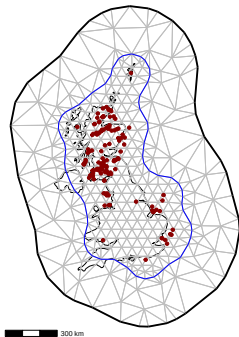
# Errors by hour

Power estimates Time Series 2025-08-10

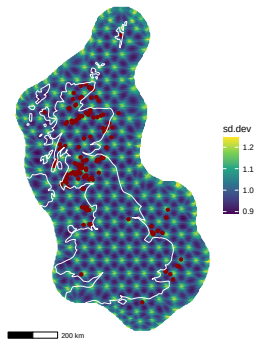


# Model Mesh

Mesh used for spatial calibration model



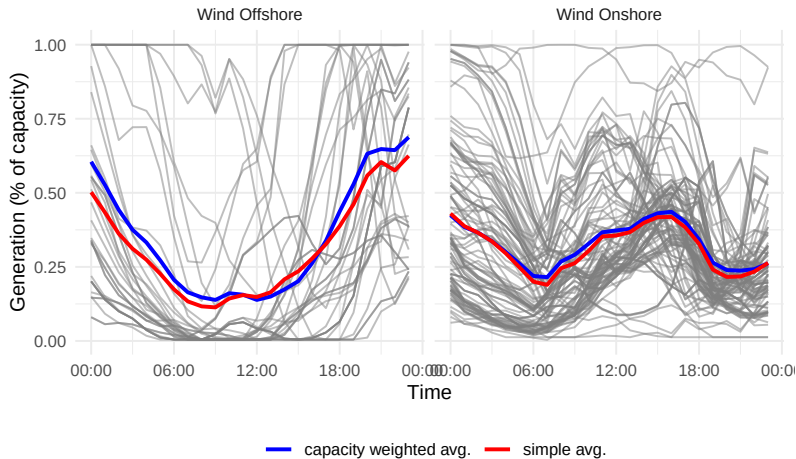
Mesh assessment



Time series of data and generic PC estimate

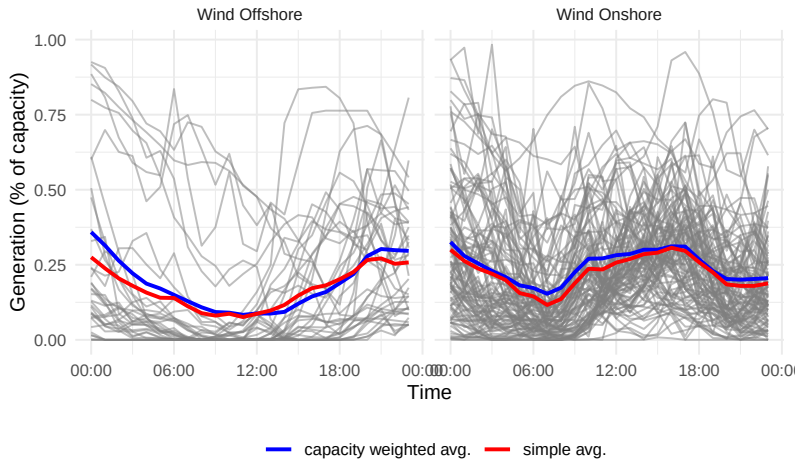
# ERA5 data and generic PC estimate

ERA5 + Generic PC Time Series 2025-08-10

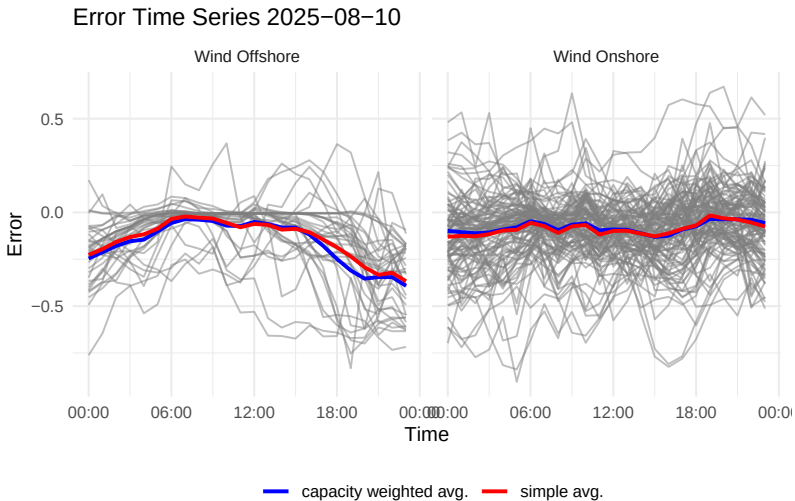


# Observed data

Observed power Time Series 2025-08-10



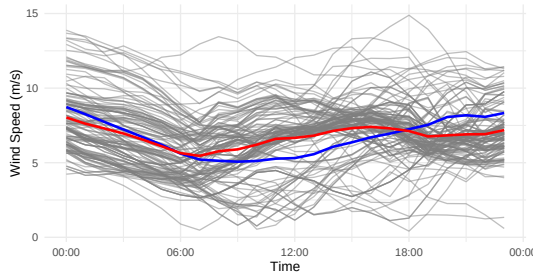
# Error of generic PC estimate



# ERA5 wind speed

## Wind speed at wind farm locations

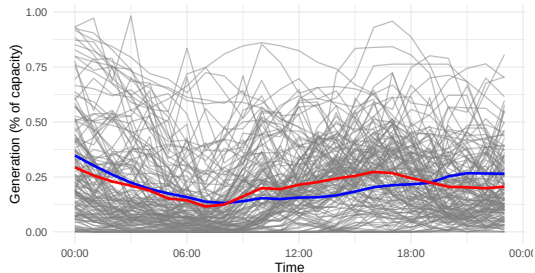
Wind Speed Time Series 2025-08-10



— capacity weighted avg. — simple avg.

## Observed generation

Observed power Time Series 2025-08-10

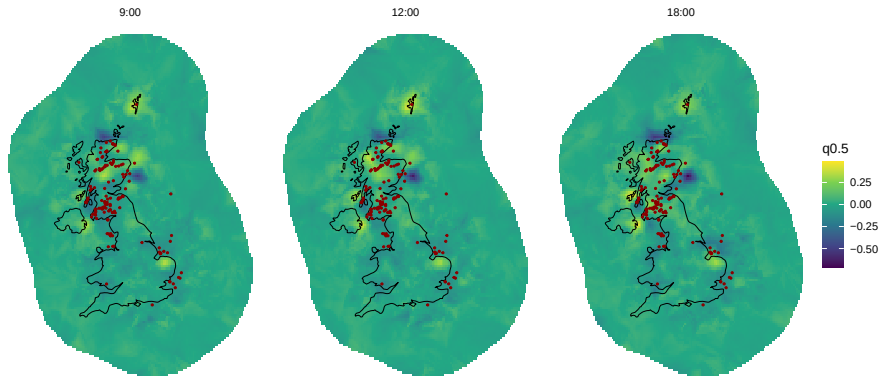


— capacity weighted avg. — simple avg.

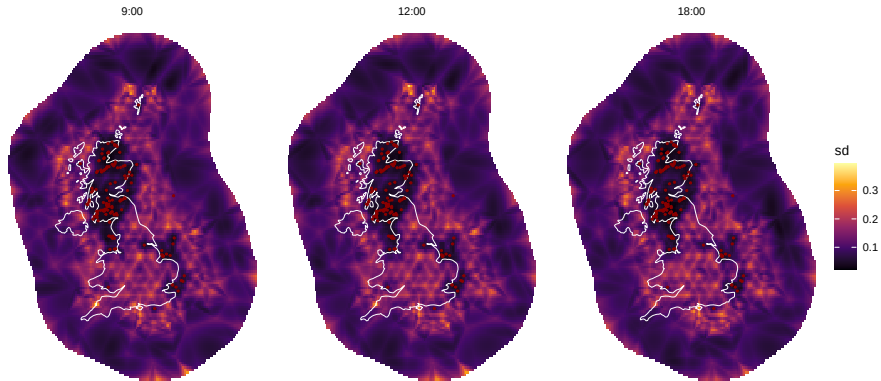
Same day previous year



# Estimated spatial field



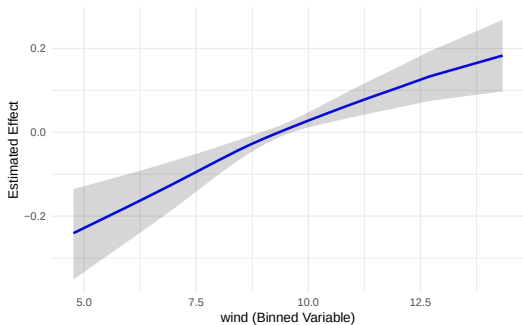
# Standard deviation of the spatial field



# Random Effects

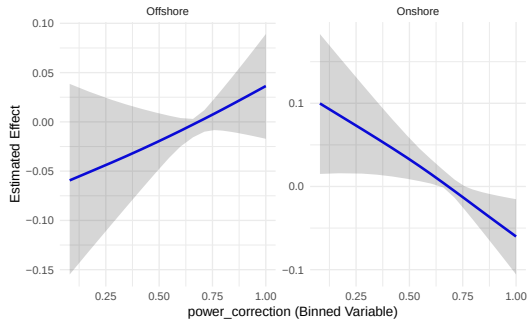
## Correction by wind speed

Estimated effect for wind



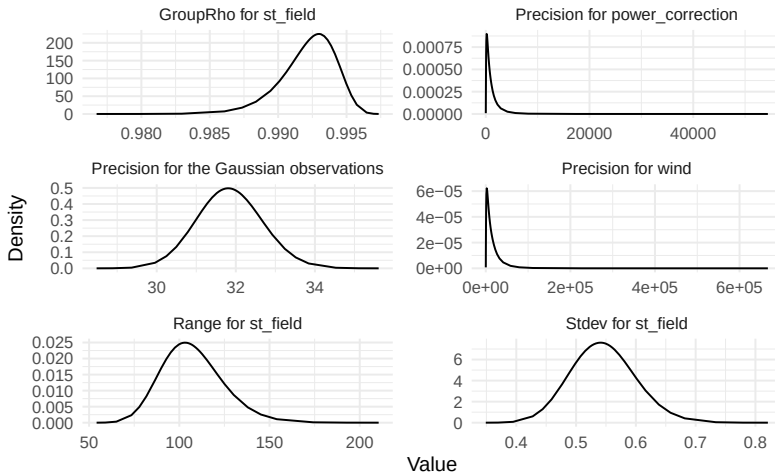
## Correction by power regime

Estimated effect for power\_correction



# Hyperparameters

## Density of Hyperparameters



## Comparison scores

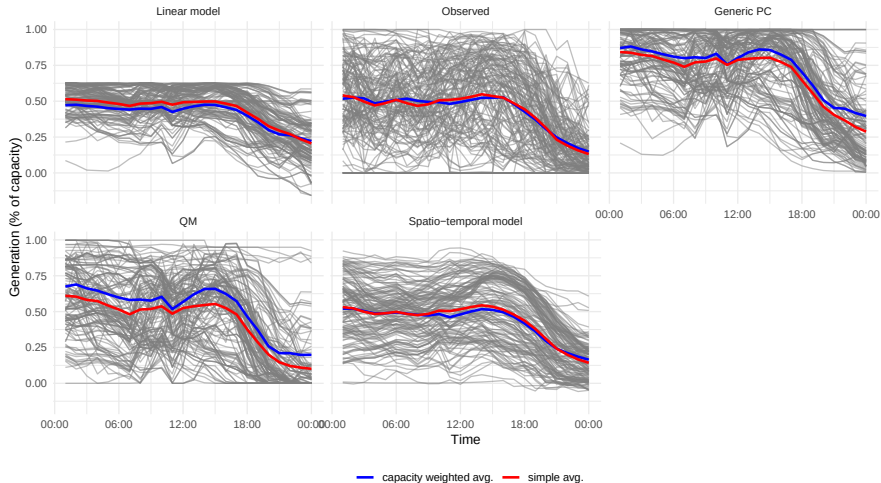
| model                 | RMSE  | MAE   | Bias      |
|-----------------------|-------|-------|-----------|
| Generic PC            | 0.385 | 0.299 | 2.50e-01  |
| QM                    | 0.314 | 0.239 | -1.23e-04 |
| Linear model          | 0.251 | 0.199 | 3.38e-17  |
| Spatio-temporal model | 0.170 | 0.125 | 1.08e-09  |

## Scores by Offshore / Onshore

| tech_typ      | model                 | RMSE  | MAE   | Bias      |
|---------------|-----------------------|-------|-------|-----------|
| Wind Offshore | Generic PC            | 0.449 | 0.369 | 3.61e-01  |
| Wind Offshore | QM                    | 0.341 | 0.236 | 1.47e-01  |
| Wind Offshore | Linear model          | 0.211 | 0.162 | 7.09e-17  |
| Wind Offshore | Spatio-temporal model | 0.098 | 0.072 | -4.90e-04 |
| Wind Onshore  | Generic PC            | 0.361 | 0.276 | 2.13e-01  |
| Wind Onshore  | QM                    | 0.305 | 0.240 | -4.88e-02 |
| Wind Onshore  | Linear model          | 0.263 | 0.211 | 2.16e-17  |
| Wind Onshore  | Spatio-temporal model | 0.187 | 0.143 | 1.62e-04  |

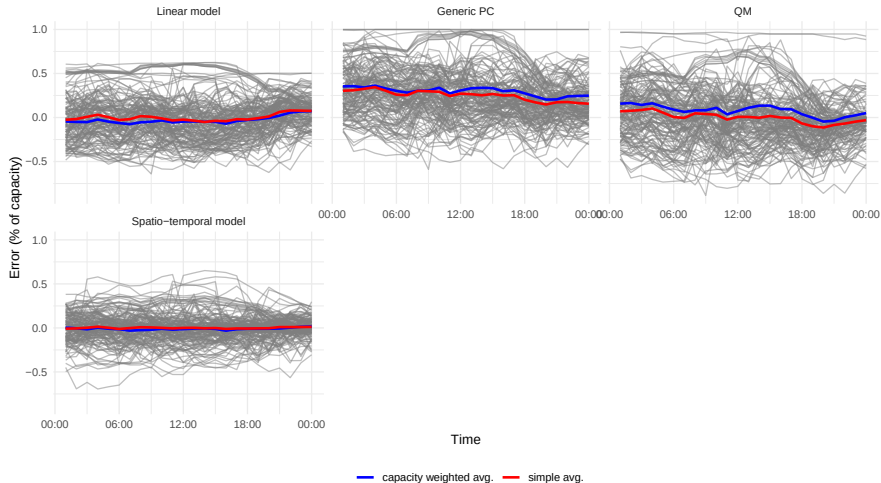
# Estimates by hour

Power estimates Time Series 2024-08-10



# Errors by hour

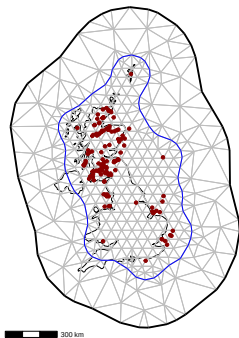
Power estimates Time Series 2024-08-10



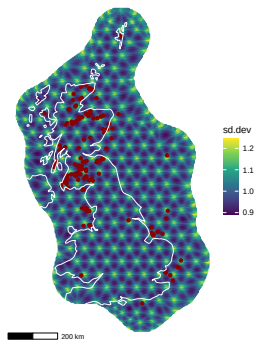


# Model Mesh

Mesh used for spatial calibration model



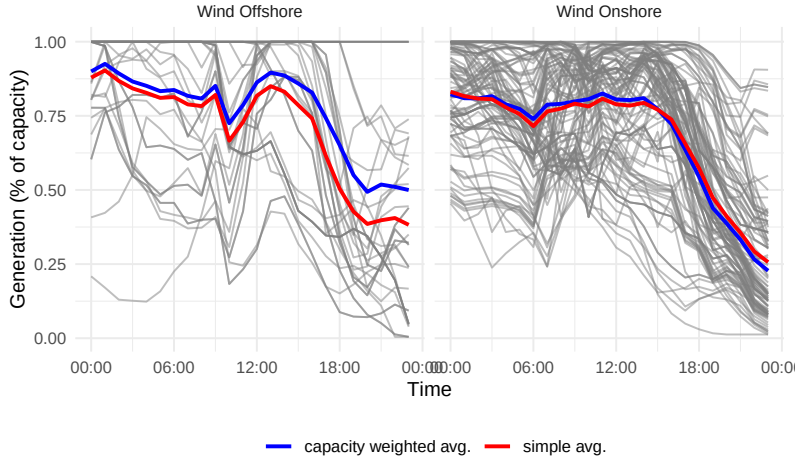
Mesh assessment



Time series of data and generic PC estimate

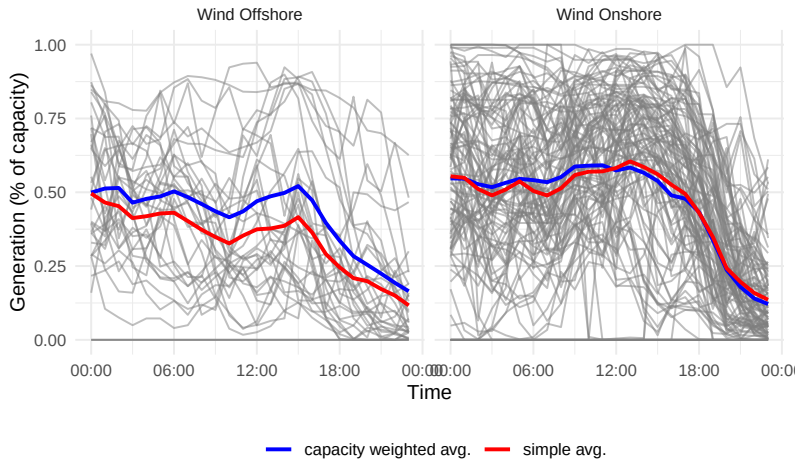
# ERA5 data and generic PC estimate

ERA5 + Generic PC Time Series 2024-08-10

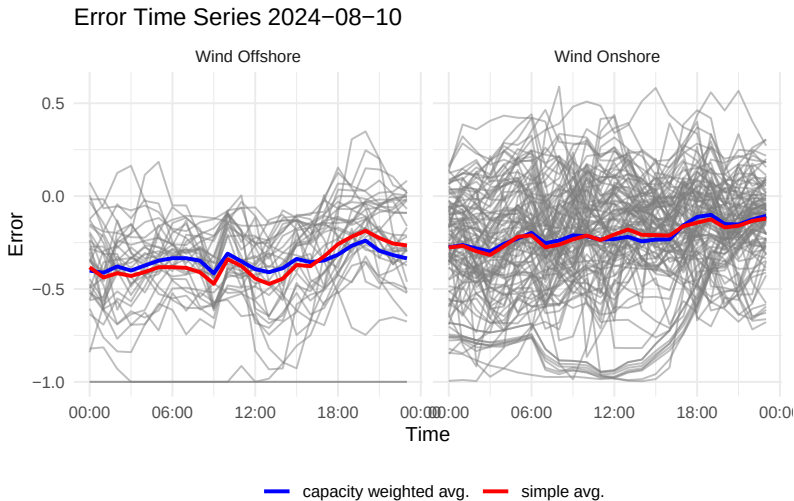


# Observed data

Observed power Time Series 2024-08-10



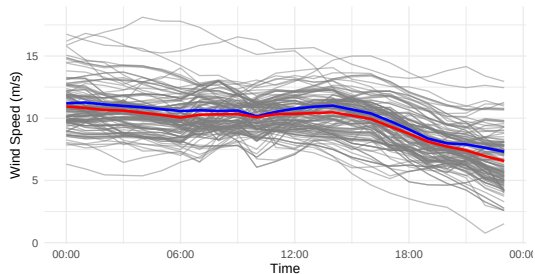
# Error of generic PC estimate



# ERA5 wind speed

## Wind speed at wind farm locations

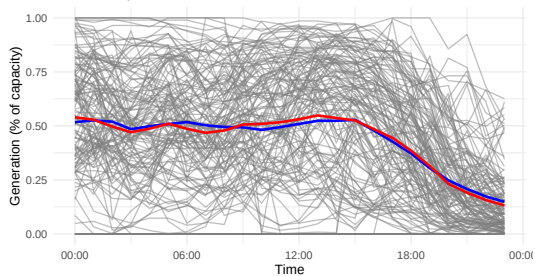
Wind Speed Time Series 2024-08-10



— capacity weighted avg. — simple avg.

## Observed generation

Observed power Time Series 2024-08-10



— capacity weighted avg. — simple avg.

## Appendix

Project Goal

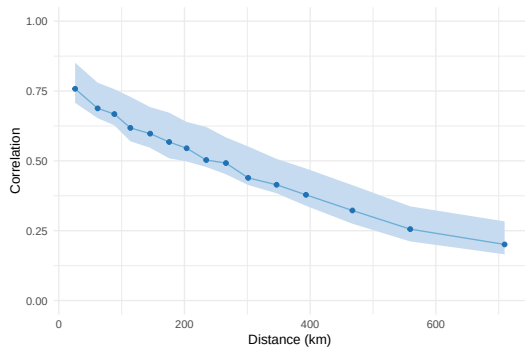


# Project Goal

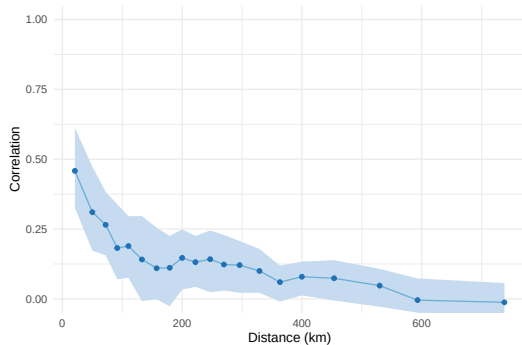
- Correct ERA5 wind power biases using observed generation data.
- Reduce error metrics (RMSE/MAE/bias).
- Ensure spatially consistent estimates that generalise out-of-sample.
- Capture low-wind events: frequency, duration, and spatial extent.

# Spatial correlation

## Correlation for Elexon generation



## Correlation for error after calibration



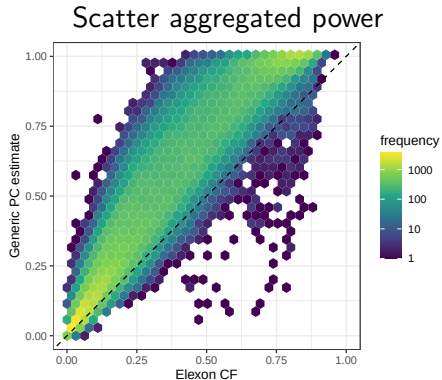
## Calibration of generic power curve estimate

# GB level aggregation

- Nearest ERA5 grid for wind speed
- Vertical interpolation using a power law
- Generic Power Curve (onshore, offshore)
- Normalised power aggregated at GB level

$$p_{\text{tot},t} = \sum_i p_{it} / \sum_i C_i$$

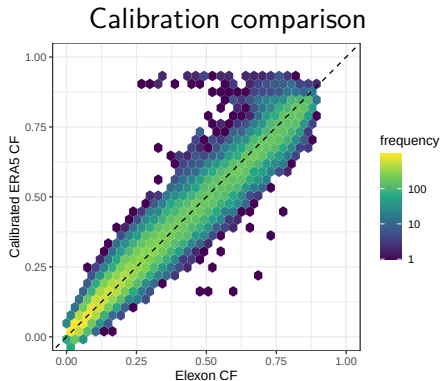
- Hourly resolution
- Potential generation (ex curtailment)



# Linear model

- Using step AIC criteria selection:
  - $k$ : Type (offshore / onshore)
  - $m$ : month,  $h$ : hour
  - $w_t$ : wind speed

$$p_t^{calib} = \alpha_{k,m,h} + \beta_{k,m} \hat{p}_t + \sum_{j=1}^3 \phi_j w_t^3$$



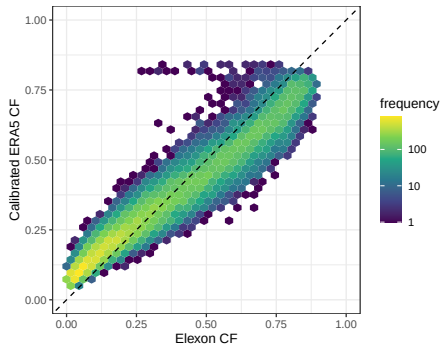
# Autoregressive error

- Using step AIC criteria selection:
  - $k$ : Type (offshore / onshore)
  - $m$ : month,  $h$ : hour
  - $w_t$ : wind speed
  - $u_t$ : AR1

$$p_t^{calib} = \alpha_0 + \sum \alpha_e + \beta_k \hat{p}_t + f_w(w_t) + u_t$$

where  $\alpha_0$  is the intercept, the  $\alpha_e$   $e \in k, m, h$  represent random effects (cyclic for month and hour),  $\beta_k$  is a random slope,  $f_w$  is smooth effect (RW2).

Calibration comparison

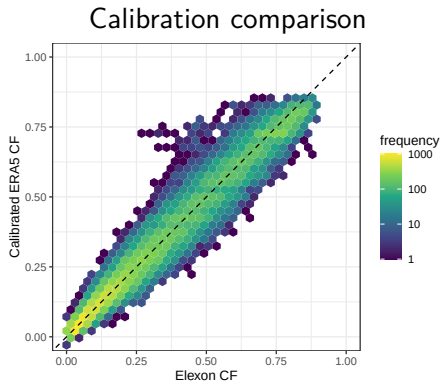


# Hierarchical Bayesian model

- Turning multiplicative corrections into smooth functions

$$p_t^{calib} = \alpha + \beta_k + \beta_m + \beta_h + s_p(p_t) + s_w(w_t)$$

where  $\beta_k$ ,  $\beta_m$ ,  $\beta_h$  are random effects and  $s_p$  and  $s_w$  are smooth functions

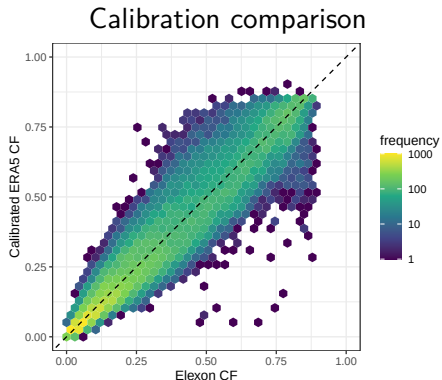


# Quantile Mapping

- Quantile mapping aims to replicate observed distribution

$$p_t^{calib} = F_{obs}(F_{mod}^{-1}(p_t))$$

where  $F_{mod}$  and  $F_{obs}$  are the CDF of the power estimate and observed values





## Comparison scores

| model          | RMSE  | MAE   | Bias      |
|----------------|-------|-------|-----------|
| Generic PC     | 0.200 | 0.164 | 1.63e-01  |
| AR inlabru     | 0.092 | 0.077 | 3.16e-04  |
| AR1            | 0.075 | 0.062 | 8.57e-05  |
| QM             | 0.075 | 0.054 | -2.60e-05 |
| Linear         | 0.052 | 0.038 | 1.39e-19  |
| Linear inlabru | 0.049 | 0.036 | -2.61e-10 |

## Power curve modelling

# Power curve modelling

## Components

- Power curve estimate from data
- Probability of zero generation
- Penalisation to make power curve resemble manufacturer's PC

Using a three-likelihood approach for power curve estimation:

- **1st likelihood:** Observed power is modelled as a smooth function of wind speed
- **2nd likelihood:** Zero inflated component
- **3rd likelihood** Penalisation towards a generic PC

# Power model overview

- 1  $\pi_{it} = \pi(w, i) = \Pr(P_{it} = 0)$  depends of wind speed and location

$$L^{(0)}(\theta^0 | \mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_t [\pi_i(w_{it}, i)^{\mathbb{1}(P_{it}=0)} \times (1 - \pi_i(w_{it}, i))^{\mathbb{1}(P_{it}>0)}]$$

- 2 Generation when  $P_{it} > 0$  also depends on wind speed / location. Learned as a smooth function from data:  $s^{\text{pc}}(w, i) \in (0, 1)$

$$s^{\text{pc}}(w_{it}, i) = \mathbb{E}(P_{it} | P_{it} > 0, w_{it}, i) = \mu_{it}^+ \\ L^{(\beta)}(\theta^{\text{pc}} | \mathbf{p}, \mathbf{w}, \mathbf{i}) = \prod_{t: P_{it} > 0} \text{Beta}(P_{it} | \mu_{it}^+, \phi),$$

where  $\theta^0$  and  $\theta^{\text{pc}}$  represents the parameters of the smooth functions,  $\pi(w, i)$  and  $s^{\text{pc}}$ , and  $\phi$  the precision of the Beta distribution.

# Penalisation component

- 3 Finally, we penalise  $s^{\text{pc}}$  to be close to our generic power curve estimate,  $\hat{P}\hat{C}_i(w)$ :  $d_{it} = |\hat{P}\hat{C}_i(w_{it}) - s^{\text{pc}}(w_{it}, i)|$  should be small.
- Or maximise the negative of  $\tau_{\text{ps}} \sum d_{it}^2$  for a certain penalisation parameter  $\tau_{\text{ps}}$ .
- Adding that penalty term is similar to assume  $d_{it} \sim N(0, 1/\tau_{\text{ps}}^{-1})$

$$\max_{\theta} \quad l^{(0)}(\theta) + l^{(\beta)}(\theta) - \tau_{\text{ps}} \sum_{i,t} d_{it}^2$$

where  $\theta = (\theta^0, \theta^{\text{pc}}, \tau_{\text{ps}})$  represents the parameters of both likelihoods, plus the penalisation parameter.

# Power curve model equations

Model is a Zero Inflated Beta (ZIB) with penalisation component:

## Zero-probability model (Bernoulli)

$$\eta_{it}^{(0)} = \alpha_{\text{zero}} + f_{\text{zero}}(w_{it}, i),$$

$$Z_{it} \mid \eta_{it}^{(0)} \sim \text{Bernoulli}(\pi_{it}), \quad \pi_{it} = \frac{1}{1 + \exp\left\{-\eta_{it}^{(0)}\right\}},$$

## Positive-output model (Beta)

$$\eta_{it}^{(\beta)} = \alpha_{\text{pc}} + f_{\text{pc}}(w_{it}, i),$$

$$P_{it}^{\text{obs}} \mid \eta_{it}^{(\beta)} \sim \text{Beta}(\mu_{it}^+, \phi), \quad \mu_{it}^+ = \frac{1}{1 + \exp\left\{-\eta_{it}^{(\beta)}\right\}},$$

## Pseudo-likelihood model (Gaussian)

$$\hat{P}C_i(w_{it}) \mid \eta_{it}^{(\beta)} \sim \mathcal{N}\left(\eta_{it}^{(\beta)}, \tau_{\text{ps}}^{-1}\right), \quad \tau_{\text{ps}} = \text{precision or penalisation}.$$

# Details on smooth components

- The smooth components  $f_{\text{zero}}(w, i)$   $f_{\text{pc}}(w, i)$  is modelled as a 1D SPDE over wind speed with a uniform correlation matrix across locations  $i = 1, \dots, I$ .

$$f(w, i) = \sum_{k=1}^I \beta_k \psi_k(w),$$

where  $\psi_k(w)$  are basis functions defined over wind speed and  $\beta_k$  are coefficients. The correlation structure of  $f(w, i)$  assumes a Matern covariance function: Let  $w_s, w_t$  represent two wind speed values in location  $i$ , and  $d = |w_s - w_t|$  their distance, then:

$$\begin{aligned} \text{Cor}(d) &= \text{Cor}(f(w_s, i), f(w_t, i)) \\ &= \frac{1}{2^{\nu-1} \Gamma(\nu)} (\kappa d)^\nu K_\nu(\kappa d), \quad \alpha = \nu + d/2, \end{aligned}$$

## Details on smooth components

- Additionally a uniform correlation matrix is assumed across locations  $j, k \in \{1, \dots, I\}$ . That is:

$$\text{Cor}(f(w, j), f(w, k)) = \rho, \quad j \neq k$$

which helps to pool information across locations, benefiting locations with less data.



# Power estimate

This leads to the following power estimate at time  $t$  and location  $i$ :

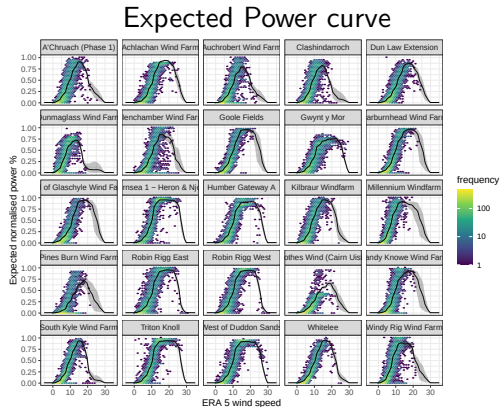
$$\hat{P}_{it} = \begin{cases} 0, & \text{with probability } \pi_{it}, \\ \mu_{it}^+, & \text{with probability } 1 - \pi_{it}. \end{cases}$$

Which leads to the expected power estimate:

$$E[P_{it}|w_{it}, i] = (1 - \pi_{it}) \times \mu_{it}^+$$

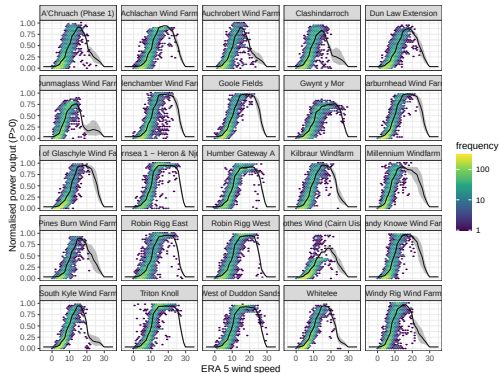
# Power curve model estimates

- Starting models with
  - 1Y of data (2024)
  - 25 wind farms
  - Hourly data
  - Excluding outages listed in REMIT

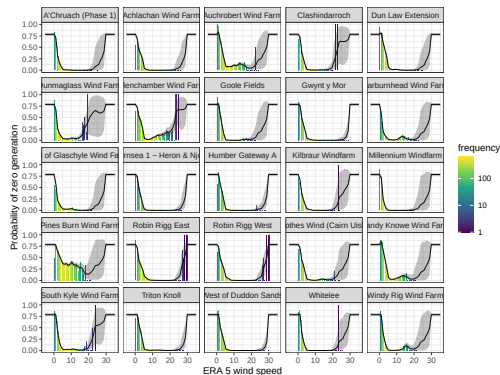


# Power curve model estimates

## Power curve estimate



## Probability of zero generation

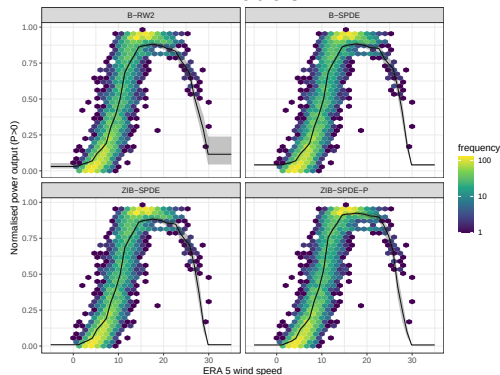


# Alternative versions

- Beta model with RW2 power curve (B-RW2)
- Beta model with 1D SPDE power curve (B-SPDE)
- ZIB model with 1D SPDE power curve (ZIB-SPDE)
- ZIB model with 1D SPDE power curve penalised (ZIB-SPDE-P)

Zero Inflated models better capture low wind power regions.

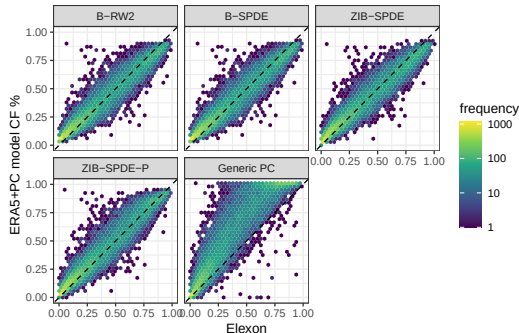
Power curve estimates under different models



# ZIB model vs observed data

- Power curve modelling highly improves the calibration
- ZIB models deal better with the tail regions
- Penalised version actually pulls estimate toward Generic PC
- We could pull estimate instead to a 5 parameter curve estimate

Power curve models vs observed data



## Error metrics

Error metrics calculated on 30% of the sampled wind farms data.

| model      | RMSE   | MAE    | Bias   |
|------------|--------|--------|--------|
| Generic PC | 0.2188 | 0.1583 | 0.1469 |
| B-RW2      | 0.0995 | 0.0723 | 0.0191 |
| B-SPDE     | 0.0994 | 0.0722 | 0.0190 |
| ZIB-SPDE   | 0.0976 | 0.0691 | 0.0103 |
| ZIB-SPDE-P | 0.1016 | 0.0694 | 0.0269 |

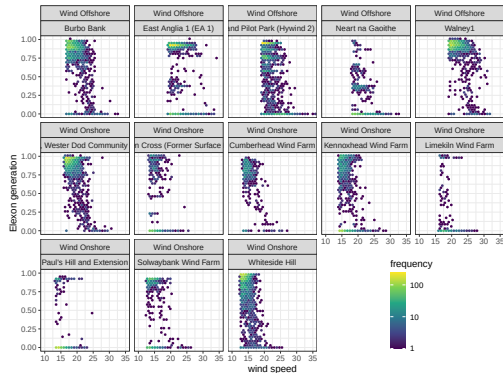
# Pooling data for cut-off estimation

- Not enough data to estimate at wind farm level
- Even using all 5 years of data (2019-2024), only a few farms show cut-off behaviour

Pooling data at high wind speeds

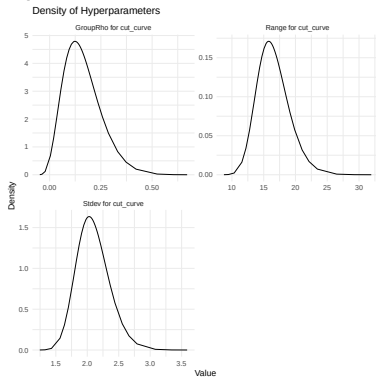
| wind speed | prop. $P=0$ | prop. $P \leq 0.02$ | n obs. |
|------------|-------------|---------------------|--------|
| [0,15]     | 0.22        | 0.23                | 3292   |
| (15,20]    | 0.13        | 0.14                | 7062   |
| (20,25]    | 0.11        | 0.13                | 5266   |
| (25,30]    | 0.43        | 0.48                | 670    |
| (30,Inf]   | 0.76        | 0.79                | 63     |

## Tail behaviour of power curve

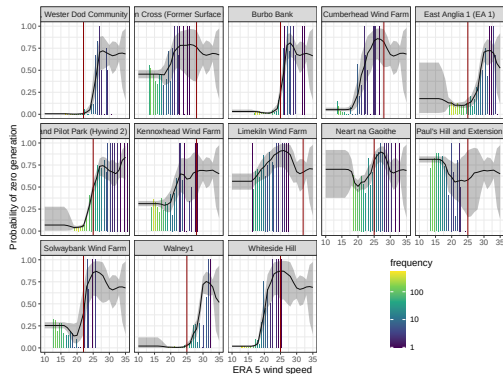


# Model to estimate cut-off by wind speed

## Hyperparameters of the cut-off model



## Probability of zero generation at high wind speeds





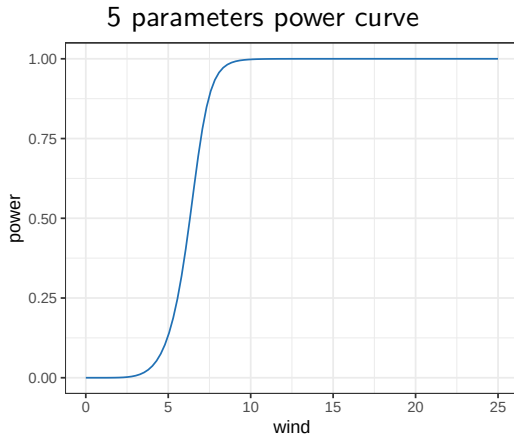
## Power Curve fitting

# Classic parametric shape

The five-parameter formula (Lydia et.al, 2013):

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

where:  $A$  and  $D$  are the upper and lower asymptotes,  $C$  controls the inflection point,  $B$  is related to the slope at inflection point, and  $G$  controls the asymmetry.



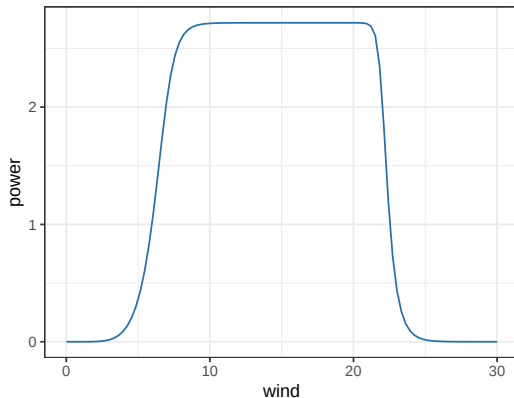
# Parametric shape for cut-off

Adding extra parameters we can model the decay after cut-off:

$$P(u) = D + \frac{A - D}{(1 + (u/C_1)^{B_1})^G (1 + (u/C_2)^{B_2})^G}$$

where:  $A$  and  $D$  are the upper and lower asymptotes,  $C_1$  is the first inflection point,  $C_2$  is related to the cut-off,  $B_1$  is related to the slope at inflection point,  $B_2$  to the slope during cut-off, and  $G$  controls the asymmetry.

7 parameters power curve



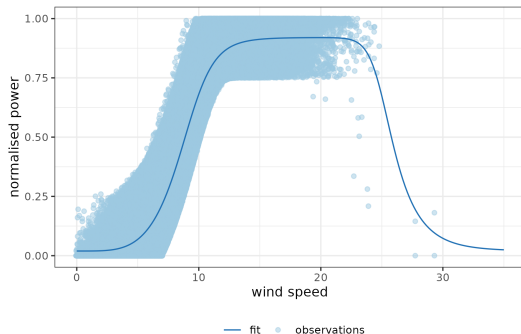
# Parametric fit for Scottish wind farms

- We can fit the parametric shape using optim
- As long as there is a reasonable amount of points showing the cut-off optim can converge

Estimates from data

| par | estimate |
|-----|----------|
| A   | 0.92     |
| B1  | -9.74    |
| C1  | 10.06    |
| D   | 0.02     |
| G   | 0.43     |
| B2  | 35.64    |
| C2  | 24.92    |

Data and power curve estimate



# Next Steps

- Spatiotemporal calibration model
- Non-stationary covariance model
- Comparison against benchmarks

# Data sources

## Wind speed

- ERA5 at wind farms
  - Hourly data
  - Spatial resolution  $0.25^\circ \times 0.25^\circ$
  - 10m and 100m heights

## Wind power

- Elexon BMU data (since 2019)
  - Half hourly data
  - Generation, curtailment, potential, capacity
  - Outage data (REMIT)
- REPD database
  - Location, turbine height, capacity

## Previous research

# Power curve modelling

## Binning method

$$P_i = \frac{1}{n_i} \sum_{j=1}^{n_i} P_{ij}$$

where:  $P_{ij}$  is the  $j$  th power observation in bin  $i$  and  $n_i$  no. of observations in bin  $i$

## Logistic

$$P(u) = a \frac{1 + m \exp(-u/\tau)}{1 + n \exp(-u/\tau)}$$

where  $a$  represents the upper asymptote,  $n, m$  shape the lower asymptote, and  $\tau$  controls the transition.



# Power curve modelling

## 5 parameter curve

$$P(u) = D + \frac{A - D}{(1 + (u/C)^B)^G}$$

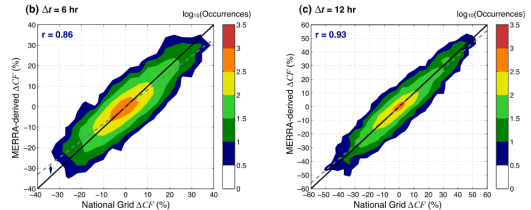
where:  $A$  and  $D$  are the upper and lower asymptotes,  $C$  is the inflection point,  $B$  the slope at inflection point, and  $G$  controls the asymmetry.

# Reanalysis data to quantify extreme wind power statistics

D. Cannon, D. Brayshaw, et. al (2015)

- MERRA wind speed validated with MIDAS
- Vertical interpolation with a logarithmic change
- Calibrated power curves based on manufacturers PC
- Use that to analyse extreme low and high levels, and ramps

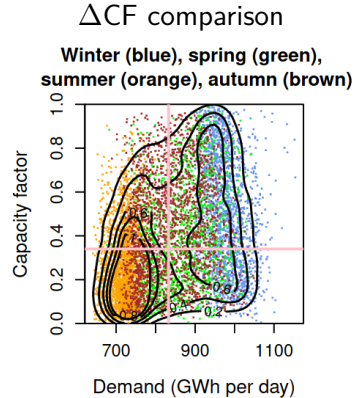
## $\Delta CF$ comparison



# Balancing energy

H. Thornton, D. Brayshaw (2020) analyse the relationship between weather, energy demand, and wind power.

- ERA Interim wind speed
- Cubic power curve with air density correction
- CF compared with GB average from other studies
- Seasonal effects on Demand and

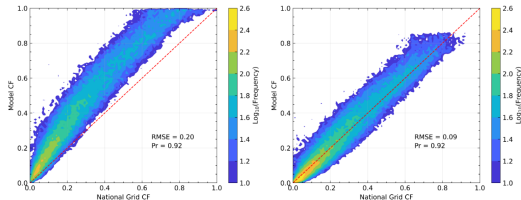


# Analysis of extreme wind droughts

Panit Potisomporn, C. Vogel (2024) perform an extreme value analysis of wind droughts in GB.

- Use ERA5 wind speeds calibrated to MIDAS with QM
- Build a ML algorithm that learns how to extrapolate wind speed from 10m to hub height
- Use a 5 parameter logistic function to model power curve
- Model energy losses with factors by type.

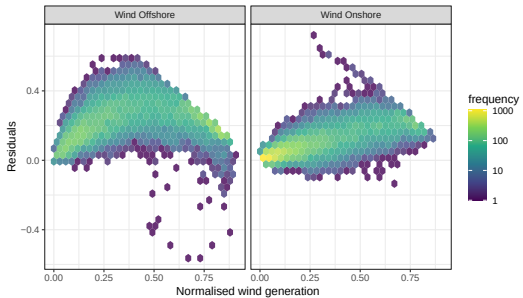
CF calibration



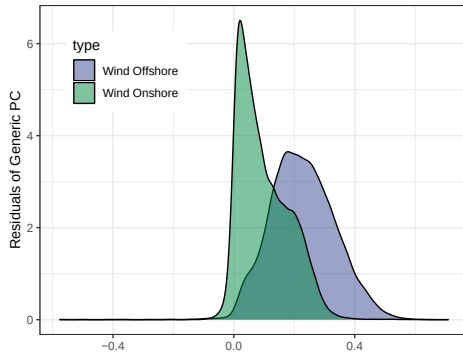
ERA5 + generic power curve error analysis

# Error by Onshore / Offshore

## Errors by observed power

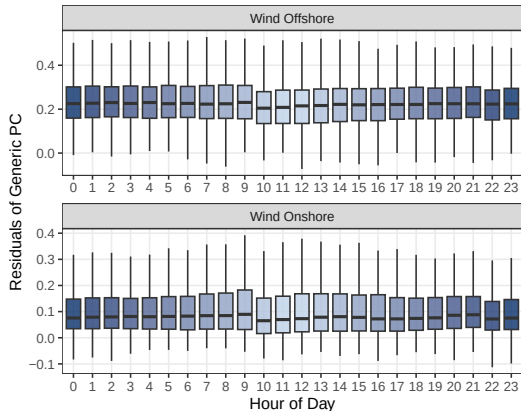


## Error density

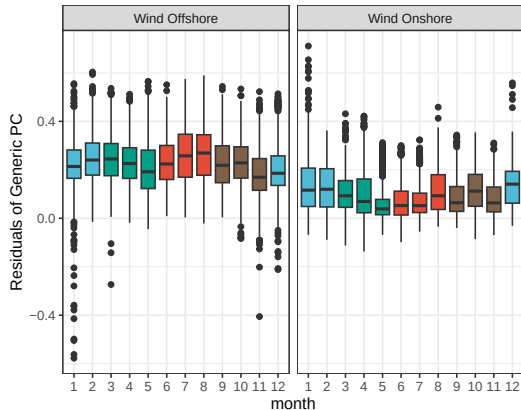


# Errors through time

## Error by hour



## Error by month



## Challenges in spatial wind calibration



# Challenges in spatial wind calibration

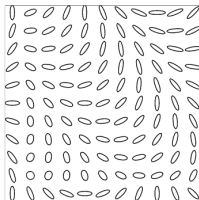
- Non-stationarity
  - (space) spatially varying coefficients
  - (wind direction) anisotropy
  - (time) regimes / storms and other weather events
- Physically informed model
  - Regularisation / penalisation
- Data dimensionality reduction
  - Aggregate time dimension (day)

Non-stationarity approaches found in the literature

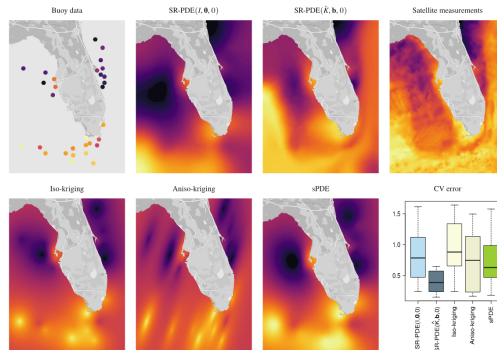
# Regression with PDE regularisation

- Tomasetto et al.(2024) model anisotropy through a physics informed regression model
- Including physical knowledge through a PDE regularisation term

Anisotropic non-stationary diffusion



## Sea Surface Temperature (SST)



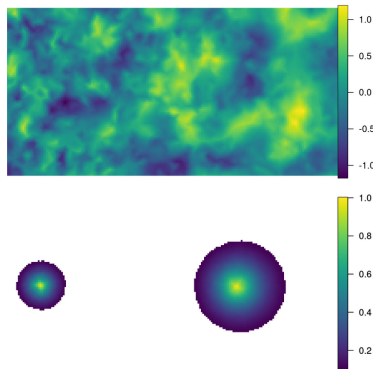
# Explanatory variables in the covariance

- Ingebrigtsen, Lindgren, and Steinsland (2014)
- Regression on ranke and precision parameters of spatial term

$$\log \tau(\mathbf{s}) = b_0^{(\tau)}(\mathbf{s}) + \sum_{k=1}^p b_k^{(\tau)}(\mathbf{s}) \theta_k, \quad (2)$$

$$\log \kappa(\mathbf{s}) = b_0^{(\kappa)}(\mathbf{s}) + \sum_{k=1}^p b_k^{(\kappa)}(\mathbf{s}) \theta_k. \quad (3)$$

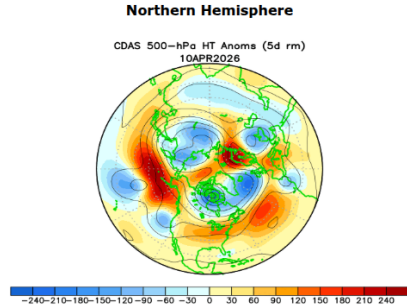
Field and correlation at two points



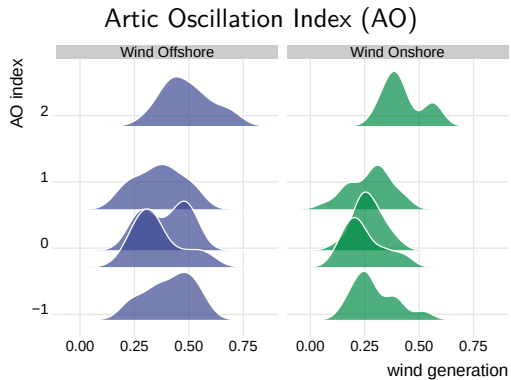
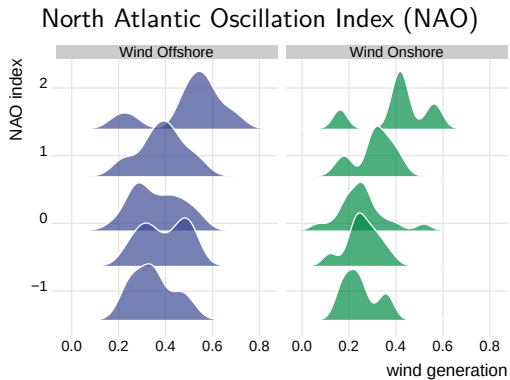
# Variables relevant to non-stationarity in wind

- Wind dynamics are non-stationary in space and time
- Weather regimes, storms, other events
- Circulation patterns (NAO, AO, EA, SCAN)
- NAO + leads to higher wind speeds UK and more variability

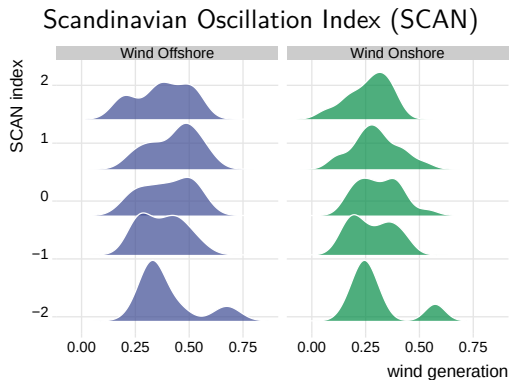
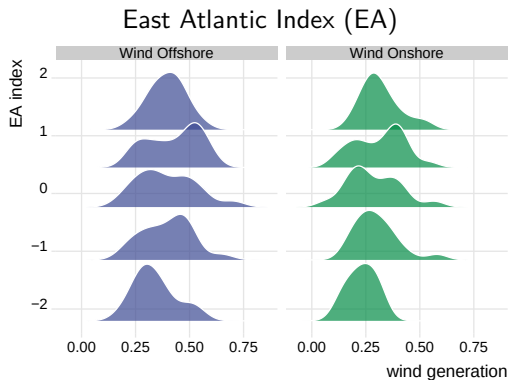
## North Atlantic Oscillation (NAO) *500-hPa Height Anomalies*



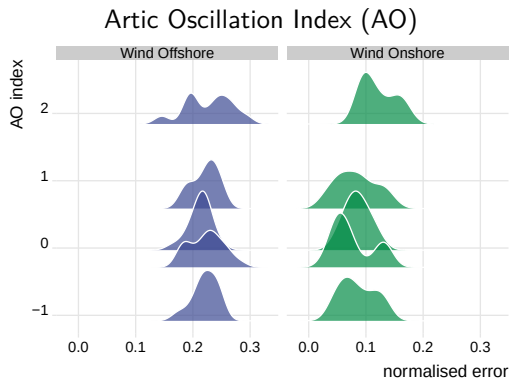
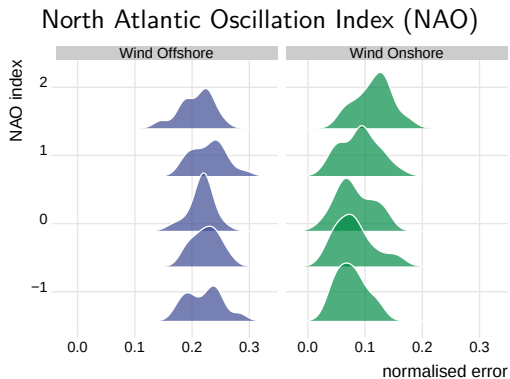
# Circulation patterns and wind power



# Circulation patterns and wind power

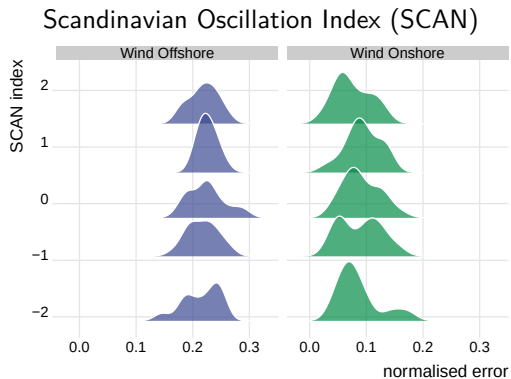
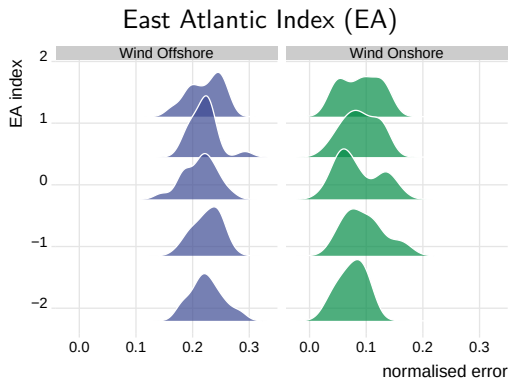


# Circulation patterns and error in wind power estimates





# Circulation patterns and error in wind power estimates



# References

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<https://doi.org/10.48683/1926.00095832>.